6	**(a)**	total resistance of circuit decreases when resistance of T decreases	B1
		current in cell increases	B1
		terminal p.d. decreases (and resistance of P is constant) so current in R decreases	B1
	(b)	p.d. of r = 1.50 – (0.200 × 6.00) = 0.30 V	C1
		current in cell = 0.30 / 0.12 = 2.5 A	C1
		resistance of thermistor = (0.200 × 6.00) / (2.5 – 0.200) = 0.52 Ω or 0.522 Ω	A1
	(c)	$I = nAvq = (N / AL)\, Avq = (N / L)\, vq$ $v = IL / Nq = (0.200 \times 0.223) / (1.10 \times 10^{20} \times 1.60 \times 10^{-19})$	C1
		$v = 2.53 \times 10^{-3}$ m s^{-1}	A1